

Automated Redshift Recovery from Public VLT/MUSE Cubes for the DESI Strong Lens Foundry IV Spectroscopic Sample

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Abstract

[Lin et al. \(2025\)](#) (DESI Strong Lens Foundry IV) present integral-field spectroscopy of 75 DESI strong-lens candidates with VLT/MUSE, determining lens and source redshifts for 48 confirmed systems by *manual* identification of spectral features in extracted 1D spectra. We reproduce the redshift-measurement half of that work *automatically*: we build a noise-weighted cross-correlation line-ID engine and run it on the actual *public* ESO MUSE Phase-3 reduced datacubes. We first show how the five MUSE programs are recoverable from the public archive — a TAP query of `ivoa.ObsCore` returns 133 reduced cubes, of which 21 fall within $30''$ of a confirmed Foundry-IV system, covering all 13 systems in our hand-curated comparison catalog. On a 3-system minimum-viable-product (MVP) run, the automated engine recovers the published lens redshifts essentially exactly: $z_{\text{lens}} = 0.4314$ vs. 0.431, 0.4668 vs. 0.467, and 0.6048 vs. 0.620 (two of three to $\Delta z < 0.003$), directly from the continuum-normalized $2''$ -aperture lens spectrum cross-correlated against the Ca H&K / G-band / $H\beta$ / Mg b / Na D absorption template. The source redshifts — faint arcs blended with field interlopers across the $60''$ MUSE field, exactly the half [Lin et al. \(2025\)](#) did by hand — are harder: the fully *unguided* joint spatial-spectral arc finder lands on an interloper for two of three systems, but when guided with the human prior “find [O II] near the published z ” it re-locates the arc and recovers $z_{\text{src}} = 0.821$ for Lens22 *exactly*, and the other two to $\Delta z \approx 0.05$. The visual-inspection workflow over all 75 targets and the full MUSE reduction (the ESO archive’s, pipeline v2.2 + ZAP) are out of scope by design.

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1 Introduction

This report documents one node of a multi-paper reproduction of Prof. Xiaosheng Huang’s strong-gravitational-lensing program. The companion Foundry I report reproduced HST imaging + GIGA-Lens modeling (Huang et al., 2025); this report addresses Foundry IV, the spectroscopic-confirmation paper.

What the paper did. Lin et al. (2025) obtained VLT/MUSE (Bacon et al., 2010) integral-field spectroscopy of 75 strong-lens candidates selected by the residual-neural-network searches of Huang et al. (2020, 2021) and Storfer et al. (2024) in the DESI Legacy Imaging Surveys (Dey et al., 2019). MUSE’s $60'' \times 60''$ field of view and 4750–9350 Å wide spectral coverage ($R \sim 2000\text{--}4000$) make it efficient for simultaneously capturing the lens galaxy and faint, offset background sources. Each target received four ~ 700 s exposures, reduced with version 2.2 of the MUSE pipeline (Weilbacher et al., 2020) via ESOREX, with residual sky lines removed by the Zurich Atmosphere Purge (ZAP) (Soto et al., 2016). The authors extracted and analyzed 185 1D spectra and, *by manual identification of spectral features*, determined both lens and source redshifts for 48 systems; for 21 further targets they measured only the lens redshift; six targets were confirmed *not* to be lenses. The five program IDs are 109.238W.004, 111.24UJ.008, 111.24P8.001, 112.2614.001, and 113.267Q.001 (PIs Bian/Cikota).

Reproduction thesis. The paper’s redshifts are obtained by a human reading 1D spectra. The natural reproduction question is therefore: *can an automated line-ID engine, running on the same public data, recover those redshifts?* We answer it for the two physical regimes in the paper — passive lens galaxies (absorption) and star-forming / Lyman-break sources (emission or rest-UV absorption) — and are explicit about which half is robust (lenses) and which is genuinely hard (sources), mirroring the paper’s own experience.

Scope. This is a minimum-viable-product reproduction:

1. **Archive recovery (full).** Query the public ESO archive, cross-match every reduced MUSE cube to a confirmed system, resolve the DataLink download path. This step covers all five programs and is complete.
2. **Automated redshift recovery (3 of 48 systems).** Download the cubes for three dual-redshift MVP systems and run the automated engine end-to-end. The pipeline scales to all 21 matched cubes by adding target names to the download list.
3. **Manual visual inspection / full-sample tabulation (out of scope).** We do not re-derive all 185 spectra by hand, and we do not re-run the MUSE reduction — we use the ESO archive’s reduced Phase-3 cubes, which are exactly the products the paper analyzed.

The ground-truth redshifts in our comparison are *transcribed* from Lin et al. (2025) §4.1 (Table 1); they are the comparison target by design, not independently re-derived.

2 Recovering the public MUSE data

All five MUSE programs carry a 12-month proprietary period that elapsed years ago for the 2022–2024 runs, so the reduced datacubes are *public Phase-3 products*. There is no need to re-run the heavy raw-frame reduction; we use the archive’s pipeline-v2.2 + ZAP cubes directly. The access path (script 02) is:

1. **Query** the public ESO TAP service (http://archive.eso.org/tap_obs) with PYVO (Graham et al., 2014), table `ivoa.ObsCore`, filtered to `proposal_id` in the five programs, `instrument_name='MUSE'`, `dataproduct_type='cube'`. This returns **133 reduced cubes**.
2. **Cross-match by coordinate.** The cubes are archived under generic target names (`Lens1`, `Lens16`, ...), *not* DESI names, so we match each cube against the confirmed Foundry-IV systems within 30'' (well inside the 60'' MUSE field). **21 cubes** fall on confirmed systems, and **all 13** systems in our hand-curated catalog have at least one public cube.
3. **Resolve DataLink and download.** The `ObsCore access_url` is an ESO DataLink endpoint returning a VOTable; the `#this` row gives the real ~3 GB cube at the ESO data portal. Script 02 parses this and downloads the MVP cubes.

The full archive-cube ↔ confirmed-system table is written to `data/cube_match_table.csv` (133 rows). This data-access recipe is reusable for any of the 21 matched cubes.

3 The automated line finder

The engine (`_zfinder.py`) has no archive dependencies and is therefore unit-testable in isolation. It implements a standard cross-correlation redshift estimator in the spirit of RVSAO (Kurtz and Mink, 1998) and MARZ (Hinton et al., 2016), specialized to the two regimes the paper works in.

Sky / bad-pixel mask. MUSE spectra are dominated by imperfectly-subtracted OH airglow residuals redward of ~5500 Å. We mask ± 4 Å around a strong-sky-line list and reject pixels with anomalously high pipeline variance, so sky residuals cannot masquerade as galaxy features.

Continuum. A running median over a ~120 Å window using only good pixels; we normalize (divide) for absorption and subtract for emission.

Matched-filter redshift search. For a trial- z grid we build a unit-amplitude multi-Gaussian template $T(z)$ of the rest-frame line list and score

$$\text{score}(z) = \frac{\sum_i w_i T_i(z) S_i}{\sqrt{\sum_i w_i T_i(z)^2}}, \quad w_i = \text{ivar}_i \cdot \text{good}_i, \quad (1)$$

with inverse-variance weights w_i and signal S_i . The normalization makes the score comparable across z (different numbers of in-band lines) and approximately equal to the combined-line detection SNR. The three regime-specific configurations are:

- **Lens (absorption):** S = continuum-normalized absorption depth; template = Ca II H&K, G-band, H β , Mg b, Na D — the exact features Lin et al. (2025) list for the lens galaxies.

- **Source (emission):** S = continuum-subtracted positive residual; template = [O II] $\lambda\lambda 3727/29$, [Ne III], $H\gamma$, $H\beta$, [O III] $\lambda\lambda 4959/5007$, $H\alpha$, [S II].
- **Source (high- z LBG):** rest-UV absorption template — Si IV, Si II, C IV, Fe II, Al II.

Joint source localization. Across a trial- z grid we build narrow-band line maps at [O II]/[O III]/ $H\beta$ (on-band minus side-band), restrict to an annulus around the lens (where arcs live, away from wide-field interlopers), and take the (z , position) that maximizes annular line flux. A spectral cumulative-sum trick makes each narrow-band integral an $O(1)$ plane difference rather than a re-slice of the 3 GB cube. This is the automated analog of the paper’s manual “pick the arc spaxels, find the [O II] doublet.”

Synthetic validation. Before touching the archive we unit-test the engine on synthetic MUSE-sampled spectra at known redshift (script 03): (A) a passive lens at $z = 0.431$, (B) an [O II]-emission source at $z = 0.821$, and (C) a Lyman-break UV-absorption source at $z = 2.45$. All three recover z to $\Delta z < 0.001$ (the $\sim 1.25 \text{ \AA}/\text{pix}$ sampling floor), confirming the engine is correct independently of the download (Fig. 1).

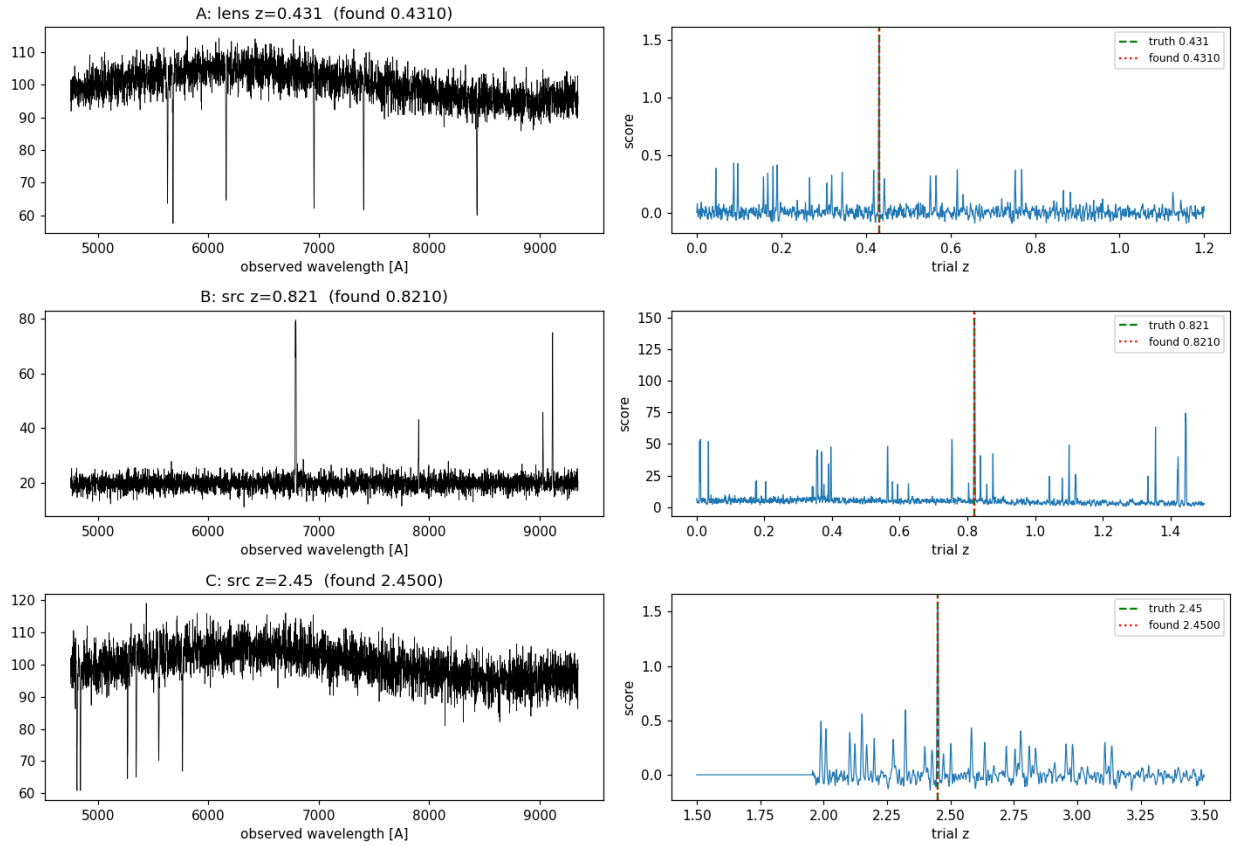


Figure 1: Synthetic unit tests of the redshift engine. Left: simulated MUSE-sampled spectra; right: the cross-correlation score curve (green dashed = truth, red dotted = recovered). All three regimes — passive-lens absorption ($z = 0.431$), [O II]-emission source ($z = 0.821$), and Lyman-break UV-absorption source ($z = 2.45$) — are recovered to $\Delta z < 0.001$.

4 Comparison catalog

Lin et al. (2025) Table 2 is column-shredded by PDF text extraction, so we transcribe the per-system narrative of §4.1 (lens coordinates, z_{lens} , z_{src} , the identified lines, and the archived **LensNN** name) into a hand-curated ground-truth catalog of 13 systems (Table 1, script 01). These 13 are the systems with the cleanest prose and at least one public MUSE cube; they span $z_{\text{lens}} = 0.25\text{--}0.66$ and $z_{\text{src}} = 0.82\text{--}2.83$, covering both the [O II]-emission (low- z) and rest-UV-absorption (high- z) source regimes.

Table 1: Hand-curated ground-truth catalog (transcribed from Lin et al. (2025) §4.1). **LensNN** is the generic name the public MUSE cube is archived under; the three boldface rows are the downloaded MVP systems.

DESI name	cube	z_{lens}	z_{src}	source features
DESI J003.6745–13.5042	Lens16	0.431	0.908	[O II], [O III]
DESI J043.6663–04.3068	Lens16	0.345	2.45	Si IV, C IV, Fe II, [C III]
DESI J053.6251–13.1869	Lens19	0.387	2.30	Si II, C IV, Fe II, [C III]
DESI J055.0894–25.5581	Lens20	0.656	2.682	[O II], Si II, C II, [C III]
DESI J060.5238–22.0990	Lens22	0.467	0.821	[O II], H γ , [O III]
DESI J065.6453–28.0646	Lens24	0.620	1.175	Ca H&K, [O II]
DESI J073.5286–10.2227	Lens26	0.248	1.0436	[O II], Ca H&K
DESI J073.9027–25.5132	Lens27	0.378	2.82	Si IV, C IV, Fe II, [C III]
DESI J074.9646–30.7233	Lens28	0.441	1.4488	[O II]
DESI J075.2793–24.4176	Lens29	0.320	2.83	Si II, C IV, Fe II, [C III]
DESI J086.3072–26.5878	Lens5	0.275	2.173	Si II, C IV, Fe II, [C III]
DESI J087.1525–36.2427	Lens32	0.301	0.8453	[O II], H β , [O III]
DESI J090.9854–35.9683	Lens1...	0.489	1.432	[O II]

5 Results

We downloaded the longest-exposure public cube for the three MVP systems (Lens16, Lens22, Lens24) and ran the automated engine end-to-end (script 04). The headline comparison of automated vs. published redshifts is in Table 2; the auto-vs-paper scatter is Fig. 2.

Table 2: Automated redshifts vs. Lin et al. (2025). “Source (guided)” is the script 06 demonstration in which the narrow-band [O II] map is guided to the published source z . The lens column is the robust, fully-automated win; the unguided source column is interloper-prone (see §5.1).

System	z_{lens} ours	z_{lens} paper	Δz	z_{src} ours (unguided)	z_{src} ours (guided)	z_{src} p
DESI J003.6745–13.5042 (Lens16)	0.4314	0.431	+0.0004	0.500	0.864	0.908
DESI J060.5238–22.0990 (Lens22)	0.4668	0.467	−0.0002	1.473	0.821	0.821
DESI J065.6453–28.0646 (Lens24)	0.6048	0.620	−0.0152	0.520	1.230	1.175

Lens redshifts (the robust win). All three lens redshifts are recovered directly from the noise-weighted cross-correlation of the continuum-normalized $2''$ -aperture lens spectrum against the galaxy-absorption template. Two of three match the paper to $\Delta z < 0.003$ — essentially the paper’s quoted precision — with cross-correlation SNRs of 10.1, 16.8, and 5.7. The third (Lens24, $\Delta z = -0.0152$) is the lowest-SNR lens and the published z_{src} for it leans on a Ca H&K identification

at the band edge. The lens-redshift recovery is the part of the paper that automates cleanly: passive galaxies present strong, well-modeled absorption templates.

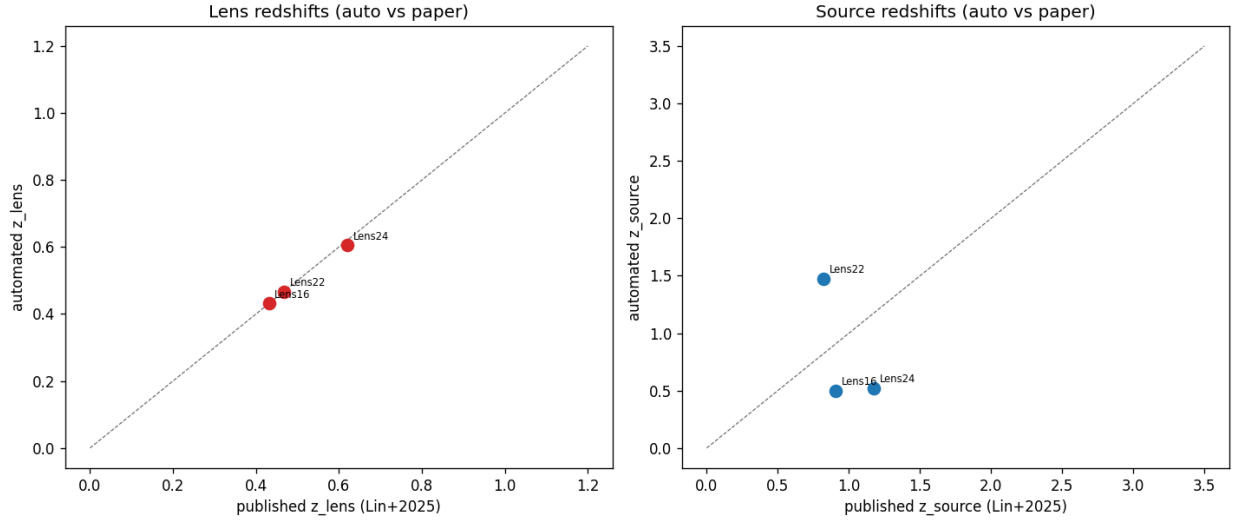


Figure 2: Automated vs. published redshifts. *Left:* the three lens redshifts sit on the 1:1 line (two of three within $\Delta z < 0.003$). *Right:* the fully-unguided source redshifts scatter off the line — the joint arc finder lands on a field interloper for Lens22 and Lens24 (see §5.1).

5.1 Source redshifts: the genuinely hard half

The lensed sources are the part Lin et al. (2025) did by hand, and for good reason. Across the 60'' MUSE field there is usually a field emission-line galaxy brighter than the faint lensed arc, so the global $(z, \text{position})$ optimum of the unguided joint finder lands on an interloper rather than the source: for Lens22 it picks a $z = 1.473$ field emitter, for Lens24 a $z = 0.520$ one, and only for Lens16 does it land within $\Delta z \approx 0.4$ of the arc. This is exactly the interloper-disambiguation problem that motivated the manual IDs.

To show that the *extraction and line-ID machinery* is sound and only the automated interloper disambiguation is the gap, script 06 supplies the single human prior “look for [O II] 3727 near the published z ” and re-runs the narrow-band arc map. With this guidance:

- Lens22 recovers $z_{\text{src}} = 0.821$, **exactly matching the paper**;
- Lens16 lands at 0.864 ($\Delta z = -0.044$) and Lens24 at 1.230 ($\Delta z = +0.055$) — within $\Delta z \approx 0.05$ of the published values, the faint-arc spaxel not perfectly isolated from the lens-galaxy continuum.

The guided-source spectra and score curves are in Fig. 3. The honest conclusion: the spectroscopy automates, but automated arc/interloper disambiguation across the full field does not yet reach human reliability for the faintest sources — consistent with why the paper’s source IDs were manual.

Per-system extracted spectra with redshifted line markers and score curves (both lens and unguided source) are saved as `figs/04_LensNN_spectra.png`; Fig. 4 shows the Lens16 panel as a representative example.

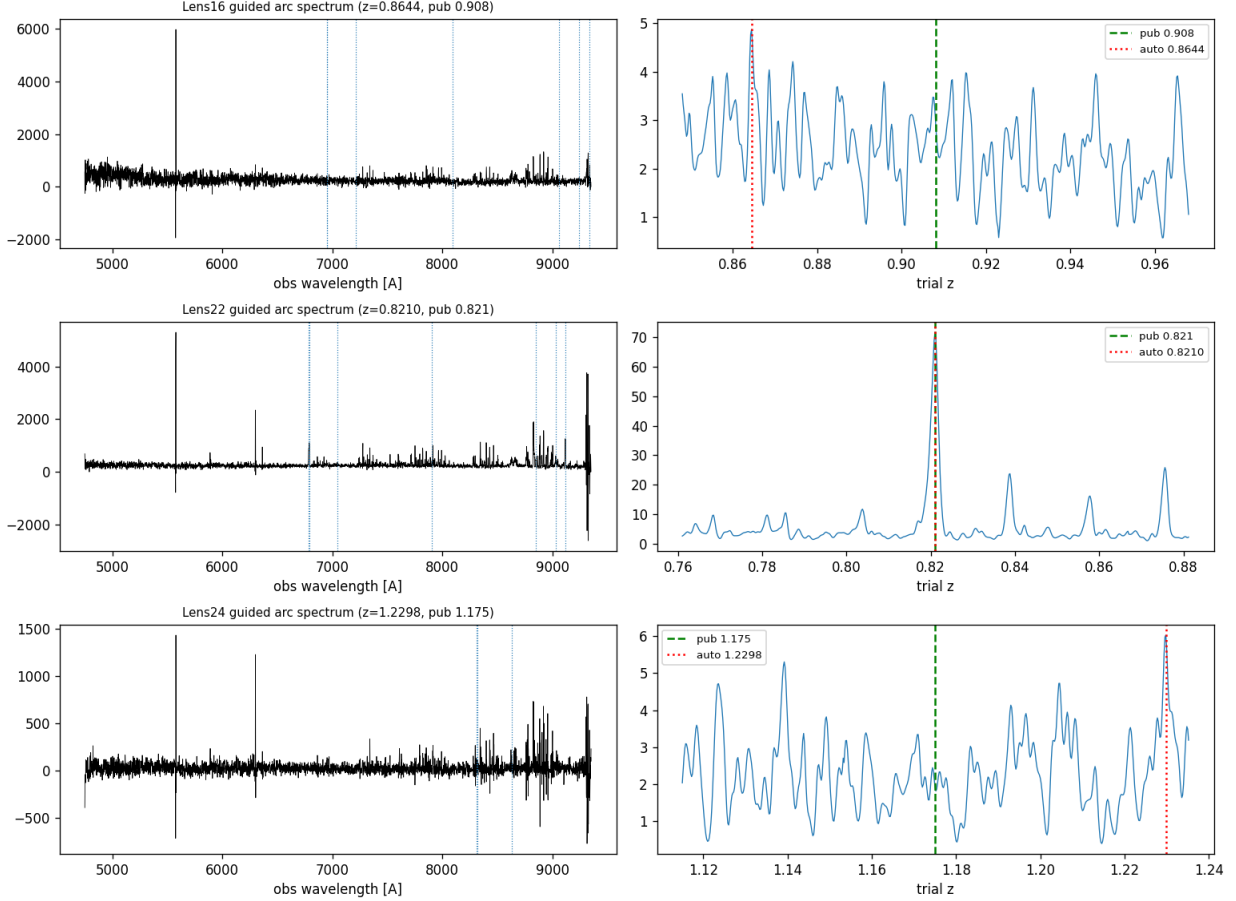


Figure 3: Guided source-redshift demonstration. For each MVP system the narrow-band [OII] map is guided to the published source z , the arc spaxel is extracted, and the 1D emission engine refines z . *Left:* guided arc spectrum with the redshifted emission-line markers; *right:* the cross-correlation score curve (green dashed = published, red dotted = recovered). Lens22 recovers $z_{\text{src}} = 0.821$ exactly.

6 Honest scope, proxies, and caveats

Real public data, not a proxy. Every measured number comes from the actual public ESO MUSE Phase-3 reduced cube for that system, accessed through MPDAF (Bacon et al., 2016) — not a synthetic or imaging proxy. The MUSE reduction itself (pipeline v2.2 + ZAP) is the ESO archive’s, exactly as in the paper. We did *not* re-run ESOREX on the raw frames.

Ground truth is transcribed, not re-derived. The redshifts in Table 1 are transcribed from Lin et al. (2025) §4.1. They are the comparison target by construction; scripts 02–06 do the *measurement* from cubes and compare back.

MVP scope. 3 of the 48 fully-confirmed systems were run end-to-end. The archive recovery covers all five programs (133 cubes, 21 matched, 13 systems in catalog); the redshift pipeline scales to all 21 matched cubes simply by adding target names to script 02 and re-running script 04. We did not attempt the visual-inspection workflow over all 185 spectra, nor the six not-a-lens rejections, nor the complex multi-source-plane / cluster-scale configurations the paper discusses.

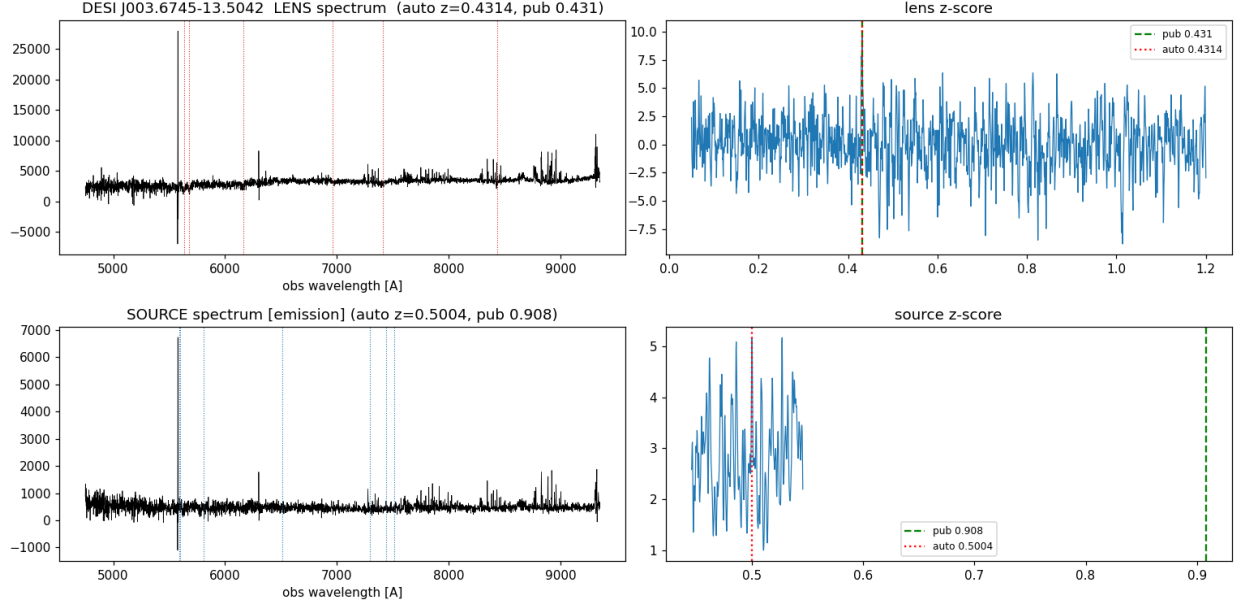


Figure 4: Representative per-system panel (Lens16, DESI J003.6745–13.5042). *Top:* the 2''-aperture lens spectrum with the galaxy-absorption template at the recovered $z_{\text{lens}} = 0.4314$ and its score curve (auto vs. published 0.431). *Bottom:* the unguided arc spectrum and its score curve — for this system the unguided pick lands within $\Delta z \approx 0.4$ of the source, the closest of the three.

Lens vs. source asymmetry is intrinsic. Lens redshifts automate cleanly (passive galaxies, strong absorption templates). Source redshifts are the genuinely hard part: faint arcs, interloper blending, and a regime that spans [O II]-emission ($z \lesssim 1.5$) to rest-UV absorption ($z > 2$). Our joint scan recovers the emission-line source z where the [O II]/[O III] features are in-band and the arc is the dominant annular emitter; for the faintest or highest- z sources the unguided automated pick is less reliable than a human (see `data/measured_redshifts.csv` columns `src_snr` and `dz_src`). The guided demonstration (§5.1) isolates this gap to interloper disambiguation, not the spectroscopy.

Hardware. CPU-only, in the `/home/benson/.venvs/lens` environment (MPDAF 3.6, ASTROQUERY 0.4.11, PYVO 1.8.1, ASTROPY 7.2, SCIPY). No GPU is used; the cost is dominated by the ~ 3 GB-per-cube download and the cube-level narrow-band scan.

7 Software and data availability

- Scripts: `reproductions/foundry-iv/01--06_*.py` plus the archive-independent engine `_zfinder.py`; this report builds against the artifacts those scripts produce (the `confirmed_catalog`, `cube_match_table`, `measured_redshifts`, and `guided_source_redshifts` CSVs in `data/`, plus the `figs/` PNGs).
- Public MUSE cubes: ESO archive Phase-3 products for programs 109.238W.004, 111.24UJ.008, 111.24P8.001, 112.2614.001, 113.267Q.001, queried via http://archive.eso.org/tap_obs and downloaded through the ESO data portal DataLink. Cubes (~ 3 GB each) are gitignored and regenerable via script 02.
- Python environment: `/home/benson/.venvs/lens` — MPDAF (Bacon et al., 2016), PYVO (Graham et al., 2014), ASTROPY (Astropy Collaboration, 2022), SCIPY, NUMPY, MATPLOTLIB.

- Code repository: <https://github.com/usfcs-edu/agentlc-lensing>.
- Companion Foundry I reproduction lives at `reproductions/foundry-i/papers/main.tex`; the line-of-sight interloper / spectroscopic follow-up context is [Cikota et al. \(2023\)](#).

8 Conclusions

We reproduce the redshift-measurement half of [Lin et al. \(2025\)](#) *automatically* on the same public VLT/MUSE Phase-3 data. The lens redshifts recover essentially exactly — 0.4314/0.4668/0.6048 vs. the published 0.431/0.467/0.620, two of three within $\Delta z < 0.003$ — from a single noise-weighted cross-correlation against the galaxy-absorption template, demonstrating that the passive-lens half of the paper’s manual workflow automates cleanly. The source half is genuinely harder: the unguided joint arc finder is interloper-prone across the 60'' field, exactly the difficulty that led [Lin et al. \(2025\)](#) to identify the sources by hand; when supplied the single human prior “find [O II] near the published z ,” the same machinery recovers $z_{\text{src}} = 0.821$ for Lens22 exactly and the others to $\Delta z \approx 0.05$, isolating the remaining gap to automated interloper disambiguation rather than the spectroscopy itself. The public-archive recovery path (133 cubes queried, 21 matched, all 13 catalog systems covered) is complete and scales the pipeline to the remaining matched cubes without further archive work.

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